



AMD DEVELOPER HACKATHON — ACT II · TRACK 3: UNICORN (OPEN INNOVATION)

CarbonPilot

An agentic AI application that analyzes the carbon, energy, and water footprint of AI workloads — and recommends greener GPU and region alternatives, reasoning live on real AMD Instinct compute.

Team: Romir Bulbule & Aarha Khanna

Repo: github.com/romirbulbule/carbonpilot

Live: frontend-seven-orpin-28.vercel.app

THE PROBLEM

AI workloads have a hidden environmental cost — and almost nobody can see it

What's missing today

- Teams choose GPUs and regions based on price and availability — carbon impact is rarely part of the decision
- Grid carbon intensity varies 15x between regions (26 gCO₂/kWh in Norway vs. 380 in Texas) but almost never factors into deployment choices
- Existing carbon-accounting tools are heavyweight enterprise ESG platforms, not something an ML engineer opens before kicking off a training run

Why it matters now

- AI training and inference workloads are growing faster than grid decarbonization
- GPU choice (AMD Instinct vs. alternatives) and region choice are two of the highest-leverage, lowest-effort levers available — but only if they're visible at decision time
- A tool that reasons about this in plain language, in real time, closes that visibility gap

WHAT WE BUILT

Describe a workload in plain English — get a live, reasoned footprint analysis

CarbonPilot takes a free-text description of an AI workload ("fine-tune a 70B model on 8 MI300X GPUs in Virginia for 24 hours"), and an agent parses it, computes energy/carbon/water impact, and reasons about greener GPU and region alternatives — all visible as a live "Thought → Tool Call → Final Answer" trace, not a black box.

ENERGY USED

144 kWh

CARBON EMISSIONS

64.5 kg CO₂e

WATER USAGE

363 liters

GREENER ALT. FOUND

97 % savings

BEYOND THE MVP

A full decision-support product, not a calculator

Plan & compare

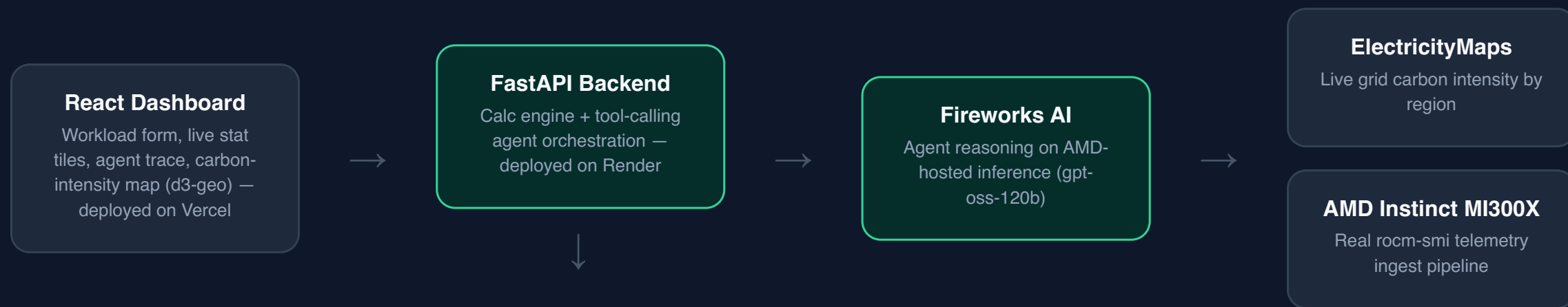
- Ranked optimization queue across every GPU × region combo — normalized by equal work delivered, not just equal wall-clock hours
- A+ to F efficiency grade, scored against every possible configuration for the same workload size
- Real-world impact equivalents — miles driven, smartphone charges, gallons of gasoline
- Electricity cost estimates alongside carbon, energy, and water

Verify & operate

- Live GPU telemetry fleet view — every reading tagged real hardware vs. simulated, so evidence is never misleading
- Interactive, drag-to-rotate globe of live grid carbon intensity by region
- Session run history with side-by-side scenario comparison and localStorage persistence
- Command palette, keyboard shortcuts, and shareable scenario deep links

HOW IT WORKS

Architecture



Every LLM call and every failure gracefully degrades — Fireworks → Claude → local regex parser — so the app is always demoable, and every response is tagged with which engine actually produced it. Live grid intensity feeds directly into the carbon math (not just the map) whenever it's available.

AMD COMPUTE USAGE

✓ Verified against real AMD Instinct hardware

This isn't a claim — here's the real telemetry

We provisioned a real MI300X instance on AMD Developer Cloud, SSH'd in, and ran `rocm-smi` directly. The readings were pushed through CarbonPilot's own live ingest pipeline — the same code path used in the product — and verified readable back out via the API.

```
$ ssh root@[mi300x-instance] "rocm-smi --showpower --showuse --showtemp --json"

{
  "card0": {
    "Temperature (Sensor junction) (C)": "38.0",
    "Current Socket Graphics Package Power (W)": "139.0",
    "GPU use (%)": "0"
  }
}
```

Instance

**rocm-7-2-4-gpu-
mi300x1-192gb-devcloud-
atl1**

Agent inference

**Fireworks AI (gpt-
oss-120b)**

Full evidence

**docs/amd-hardware-
evidence.md in repo**

Integrity

**Simulated telemetry is
tagged & labeled — never
shown as real**

WHAT'S NEXT

Roadmap beyond the hackathon

Rack/server-level oversight (Tier 2)

Real server power via IPMI/Redfish and PDU metering, for teams running their own hardware rather than cloud instances.

Cloud provider carbon APIs

AWS/GCP/Azure emissions exports as an additional live data source, account-scoped, for teams already on those platforms.

Shared team accounts

Sync run history and decision logs to a backend database instead of per-browser localStorage, so a whole team can compare scenarios together.

THANK YOU

Try CarbonPilot

GitHub

`github.com/romirbulbule/carbonpilot`

Live demo

`frontend-seven-orpin-28.vercel.app`

Team

Romir Bulbule & Aarha Khanna